When mutualism turns costly: abiotic stress shifts endophyte benefits in cool-season grasses

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Abstract

Beneficial interactions between plants and fungi are ubiquitous and foundational to terrestrial ecosystems. Plant-fungal mutualisms are expected to strengthen under environmental stress, yet the relative importance of abiotic and biotic factors in eliciting mutualistic benefits remains unclear. We studied how the demographic effects of seed-transmitted fungal endophytes on native three grass host species varied in response to abiotic and biotic factors using geographically distributed field experiments along an aridity gradient, cross with herbivore access or exclusion. Vital rate models showed that all components of host fitness (survival, growth and fertility) declined with increasing, indicating reduced performance under abiotic stress. Declines were more pronounced in individuals hosting endophytes, suggesting that interactions that are beneficial under benign conditions can become costly under abiotic stress. On the other hand, herbivory had no detectable effect on host demography or on the demographic effects of hosting endophytes. Our results indicate that abiotic stress was the primary determinant of interaction outcomes between plants and fungal symbionts, but in the opposite direction as predicted by theory and prior work. These results highlight that symbiotic fungi can become liabilities under stress, with implications for the dynamics of plant-fungal mutualisms under global change.

Introduction

Plant-microbe symbioses are widespread and ecologically important. These interactions are famously context-dependent, where the direction and strength of the interaction outcome depend on the environment in which it occurs (Bronstein, 1994; Hoeksema and Bruna, 2015). For example, plant-microbe interactions that are beneficial under stressful conditions may become neutral or parasitic under benign conditions (Giauque et al., 2019). Variation in the nature of biotic and abiotic stress is a key driver of these context-dependent outcomes.

Microbial symbioses can substantially enhance plant resilience to environmental stress. Under biotic stress, such as herbivory, microbial symbiosis can benefit host plants by facilitating the production of secondary compounds that deter feeding or exert direct toxicity, thereby reducing herbivore growth, survival, and oviposition (Atala et al., 2022; Bastias et al., 2017; Vega, 2008). Similarly, under abiotic stress, such as low precipitation, nutrient limitation, or high temperature, symbiotic microbes can enhance host tolerance through a variety of mechanisms (Afkhami and Rudgers, 2008; Clay and Schardl, 2002). Symbionts including mycorrhizal fungi, rhizobia, and endophytic bacteria can improve water and nutrient acquisition by increasing root surface area, facilitating nutrient mobilization, or promoting nitrogen fixation, allowing plants to maintain growth and metabolic function under drought or poor soil conditions (Amijee et al., 1989). In addition, many symbiotic microbes stimulate the production of antioxidant enzymes, such as superoxide dismutase and catalase, which reduce oxidative damage caused by stressors like drought, salinity, and heat (Ruiz-lazona et al., 1996). By improving resource acquisition, modulating plant physiology, and enhancing stress-responsive metabolism, these symbiotic interactions help host plants buffer against environmental stochasticity and extreme conditions across a wide range of ecosystems (Fowler et al., 2023).

Although microbial symbioses often enhance plant performance, they can also impose substantial costs, particularly in resource-limited environments. Maintaining symbionts requires hosts to allocate valuable carbon and nutrients to sustain their microbial partners. For instance, mycorrhizal fungi may require up to 20% of photosynthetically fixed carbon (Soudzilovskaia et al., 2015; Thirkell et al., 2020), and under abiotic stress, this investment can outweigh the benefits of the symbiosis (Bronstein, 2001; Johnson et al., 1997), reducing host survival, growth, reproduction, or competitive ability (Klironomos, 2003). Similarly, vertically transmitted endophytes can shift from mutualistic to antagonistic interactions when they produce stromata that abort host inflorescences or impose energetic costs (Saikkonen et al., 2004). Experimental studies further show that under low nutrient or water conditions, endophyte-infected grasses may produce fewer tillers and lower total biomass compared to uninfected plants, reflecting the costs of sharing limited resources (Ahlholm et al., 2002).

Despite growing interest in the role of symbiosis in shaping host performance, few studies have tested under natural conditions how abiotic and biotic stressors synergistically influence the demographic effects of symbionts along environmental gradients. Most existing work has been conducted in controlled or common garden experiments, where, for example, herbivory but not drought reduced plant growth, and no interaction between the two stressors was observed (Emery et al., 2010). Studying the combined effects of abiotic and biotic stressors under natural conditions is particularly important in the context of global change (Louthan et al., 2015), as shifting climate patterns and altered herbivore pressures may change the balance of costs and benefits in plant–microbe interactions, potentially influencing species distributions and ecosystem function. Here, we address this gap by examining the additive effects of biotic and abiotic stress on grass-endophyte symbioses in native populations.

Grass endophytes, especially those vertically transmitted through seeds, provide a tractable system for studying symbiosis, as infection status (symbiont-free vs. symbiont-infected) can be experimentally manipulated. A well-studied example is the association between cool-season grasses and *Epichloë* fungi (Clay et al., 2005; Oberhofer et al., 2014; Saikkonen et al., 2010; Schardl et al., 2004). Endophyte-infected grasses often perform better than endophyte free grasses under water limitation, although benefits can vary with host, endophyte species, or genotype (Saikkonen et al., 1998). Infection frequency is also influenced by herbivory and environmental conditions, including long-term variation in precipitation and temperature (Clay et al., 2005; Fowler et al., 2025; Rudgers et al., 2016). Even when endophytes do not enhance survival under stress, they may still boost host fitness by promoting growth and reproduction (Yule et al., 2013). These characteristics make grass–*Epichloë* symbioses an ideal system for investigating how abiotic stressors, such as drought, interact with biotic stressors, like herbivory, to determine the outcome of symbiosis in cool-season grasses.

Working across an aridity gradient in the south-central US, we investigated how endophyte symbiosis influences host-plant demography under varying abiotic stress. This region provides an ideal system to study biologically realistic stress gradients because it spans a strong east–west precipitation gradient, with mesic conditions in the east transitioning to increasingly arid conditions in the west. At each site along the gradient we also tested herbivory as a biotic stressor to evaluate its potential interactive effects with abiotic conditions. Specifically, we studied the symbiotic association between three native cool-season grass species (*Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*) and their vertically transmitted fungal symbionts (*Epichloë* spp.). Here we address two questions : (i) under combined abiotic and biotic stress, do endophytes provide fitness benefits or impose costs on their host plants? (ii) which factors (abiotic vs. biotic) influence whether the grass-endophyte interaction benefits or harms the host plant? We predicted that endophytes will enhance host performance under low precipitation by mitigating drought stress. These benefits are expected to be smaller in fenced plots with reduced herbivory, where the contribution of symbionts to host performance is less critical. We further predicted that the grass–endophyte symbiosis will be less mutualistic under high stress conditions, particularly when herbivores presence adds biotic stress. While both types of stress are expected to influence the outcome, we anticipated that abiotic stress will be the primary driver of mutualistic benefits, with biotic stress modulating the magnitude of those benefits.

Materials and methods

Study system

Our study focused on three cool-season perennial grasses native to woodland and prairie habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a small, short-lived perennial that germinates from autumn to late winter and typically flowers between March and July. *E. virginicus* is a larger, long-lived species that begins flowering as early as March or April and continues throughout the summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers from late spring through summer. All three species host *Epichloë* fungal endophytes, which are primarily transmitted vertically through seeds. *Agrostis hyemalis* is associated with *Epichloë amarillans* (White Jr, 1994), *E. virginicus* typically hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* can also harbor endophyte taxa capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). However, horizontal transmission is rare, influenced by environmental and genetic factors, and was not observed in our study system.

To examine the effects of endophytes on host species across a stress gradient, we established eight plots (1.5 m $\times$ 1.5 m) at each of seven common garden along the aridity gradient from Louisiana and Texas (Fig. 1A-D). Each plot contained 15 individuals of a single species, with half of the plots protected from herbivores using exclusion fences to assess the interactive effects of endophytes and herbivory (Fig. 1E). In summer–fall 2021, seeds were collected from naturally symbiotic source populations and either heat-treated to remove endophytes or left untreated, producing symbiont-free (E^{-}) and symbiotic (E^{+}) plants from the same genetic lineages.

Source populations and symbiont removal

Plants used in the common garden experiment were derived from natural (source) populations throughout the natural distribution of the species (Fig. 1,Table S1). Seeds were collected from these source populations, and some were heat-treated at 60°C for approximately five days (120 hours) to produce endophyte-negative plants (E^{-}), a method that eliminates endophytes without affecting seed viability. All seeds, both heat-treated and non-heat-treated, were planted in the Rice University greenhouse, where seedlings were fertilized every two weeks. Seedlings were then vegetatively propagated to produce enough individuals for the experiment. Before field planting, we confirmed endophyte status (E^{+} or E^{-}) of all seedlings using either microscopy or an immunoblot assay. Leaf tissues were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify fungal hyphae. The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of *Epichloë* spp. and a chromagen to visually indicate presence or absence (Koh et al., 2006). Both methods yield similar detection rates.

Common garden experimental design and climate data

We conducted our common experiment across spanning the precipitation gradient from Louisiana to central Texas, USA (Fig. 1, Table S-1). To characterize the abiotic stress gradient, we collected daily temperature and precipitation data from PRISM (Hart and Bell, 2015). These daily values were used to calculate the mean temperature (°C) and cumulative precipitation (mm) for each year, spanning from the time plants were transplanted to when demographic data were collected. During our study period (2023-2025) realized precipitation at these sites corresponded well to 30-year normals, with wetter eastern sites and drier western sites. Mean annual temperature did not differ significantly across sites (Fig. S1), so we focus on precipitation as the primary abiotic factor that varied across sites. This gradient also overlaps and exceeds the natural distributions of our focal species, ensuring realistic exposure to climatic and biotic conditions (Fig. 1A,B, C).

In each common garden, we installed eight plots per species. Plots were randomly assigned one of four initial endophyte frequency treatments (80%, 60%, 40%, or 20%) and one of two herbivore access treatments (fenced to exclude herbivores or unfenced to allow access). In the fenced plots, insecticides were applied periodically to eliminate small herbivorous insects, as *Agrostis hyemalis*, *Epichloë amarillans*, and *Poa autumnalis* are naturally subject to herbivory by both insects and larger mammals. To ensure that our herbivory treatment reflected the level of herbivory experienced by individual plants, we counted the number of damaged plants per plot and estimated the proportion of damaged plants per site and herbivore treatment. Our results confirmed that the herbivory treatment was effective (Fig. 1F).

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May, and *Elymus virginicus* was censused in June of 2023, 2024, and 2025. For each individual, plant survival was recorded as alive or dead and, for surviving plants, size was measured as the number of tillers. Growth was quantified as the log ratio of the number of tillers between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$) for plants that were alive in both years. We recorded the number of inflorescences per plant for all three species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*.

Assessing whether endophytes confer benefits or impose costs under abiotic and biotic stress

To assess whether endophytes confer benefits or impose costs under abiotic and biotic stress, we first developed two candidate models for each vital rate: one quadratic and one monotonic. All models shared the same hierarchical structure and were applied to survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, the number of inflorescences with a zero-inflated negative binomial distribution, and the number of spikelets with a negative binomial distribution. We modeled all vital rates (survival, growth, and fertility) using species-specific fixed effects for symbiosis (endophyte presence), precipitation, and herbivore access, including all relevant two- and three-way interactions. Each observation i corresponds to a plant in a given year; individuals can appear in multiple years. For simplicity, we omit the full subscript notation for species, plot, population, and site-year, which is explicitly included in the Stan model. We present the quadratic model here (Eq. 1) because it encompasses the linear model as a special case.

$$\begin{aligned}\mu_i = {} & \beta_0[Sp_i] + \beta_P[Sp_i]P_i + \beta_{P^2}[Sp_i]P_i^2 + \beta_{endo}[Sp_i]endo_i + \beta_{herb}[Sp_i]herb_i \\ & + \beta_{endo\times P}[Sp_i]endo_iP_i + \beta_{endo\times P^2}[Sp_i]endo_iP_i^2 + \beta_{herb\times P}[Sp_i]herb_iP_i \\ & + \beta_{herb\times endo}[Sp_i]herb_iendo_i + \beta_{endo\times herb\times P}[Sp_i]endo_iherb_iP_i \\ & + \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i],Sp_i} + \rho_{\text{plot}[i],Sp_i}.\end{aligned} \tag{1}$$

Here, P_i is total precipitation (mm) cumulated over the 12 months preceding the census, $endo_i$ indicates endophyte presence, and $herb_i$ indicates herbivore access. Random effects account for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity, $\omega_{\text{source}[i],Sp_i}$ captures variation associated with the provenance of transplants, and $\rho_{\text{plot}[i],Sp_i}$ captures plot-to-plot variation. Each species has its own random effect for each site, source, and plot, while the variance parameters (σ_{site}, σ_{source}, σ_{plot}) are shared across species. This hierarchical structure allows the model to "borrow strength" across species, improving estimates of fixed effects while still capturing species-specific responses to environmental gradient. We confirmed that models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles using a posterior predictive check (Fig. S2).

Secondly, we compared quadratic and monotonic candidate models using leave-one-out cross-validation (LOOCV) to identify the best-supported model for each vital rate (Vehtari et al., 2017). LOOCV combines validation and training by iteratively holding out one observation as a test set while fitting the model on the remaining $n-1$ observations. This process is repeated n times, once for each observation, resulting in n fitted models (Silva and Zanella, 2024). The overall test error estimate is then obtained by averaging the prediction errors across all n models (Eq. 2) for each vital rate.

$$CV_n = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

Although we evaluated quadratic models, LOOCV showed that they provided minimal improvement over monotonic models (Table S-2). Therefore, all subsequent analyses used the monotonic model, which captures the observed patterns while maintaining model simplicity.

Finally, for each vital rate, precipitation was summarized at key quantiles (10th, 25th, 50th, 75th, and 90th percentiles) to represent the range of site-level abiotic stress. For each species (*Agrostis hyemalis*, *Elymus virginicus*, *Poa autumnalis*) and herbivory treatment (Fenced or Unfenced), we computed posterior predictions of survival probability for individuals with endophytes (E^+) and without endophytes (E^-) using the best Bayesian model outputs (Eq. 1). The difference ($\Delta = E^+ - E^-$) was calculated for each posterior sample to estimate median effects and 90% credible intervals. We also computed the posterior probability that $\Delta > 0$, representing the likelihood that endophytes confer benefits on vital rates under the given climate and herbivory conditions.

Drivers of endophyte effects on cool-season grass demography

To identify the main drivers of endophyte-mediated effects, we fitted three hierarchical Bayesian candidate models for each vital rate (survival, growth, inflorescence, and spikelet production(Supplementary Methods S.0.2). All models included endophyte status as a baseline predictor and differed in whether they incorporated interactions with precipitation (Endophyte $\times$ Climate), herbivory (Endophyte $\times$ Herbivory), or both factors including the three-way interaction (Endophyte $\times$ Climate $\times$ Herbivory). Random effects for site-year ($\phi_{\text{site}[i],\text{Sp}_i}$), source population ($\omega_{\text{source}[i],\text{Sp}_i}$), and plot ($\rho_{\text{plot}[i],\text{Sp}_i}$) captured hierarchical variation while allowing species-specific responses, as described in the previous section. Candidate models were compared using leave-one-out cross-validation (LOO-CV) (Vehtari et al., 2017), which estimates predictive accuracy by iteratively holding out each observation. Differences in expected log predictive density (ΔELPD) were used to rank models, with standard errors quantifying uncertainty.

All models were implemented in R (R Core Team, 2023) using Stan via the `rstan` interface (Stan Development Team, 2024).

Results

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Across fitness components, *Epichloë* endophytes generally enhanced host performance under wetter conditions, while their effects were neutral or occasionally costly under drier conditions or in the presence of herbivores. Benefits were most apparent when herbivores were excluded.

For survival, endophyte effects were mostly neutral or slightly negative at low precipitation, but became increasingly positive with wetter conditions, particularly in unfenced plots (Fig. 2; Fig. S-1). In *Agrostis hyemalis* and *Elymus virginicus*, median $\Delta(E^{+} - E^{-})$ ranged from approximately -0.03 to 0.04 under low precipitation in unfenced plots ($\Pr(\Delta > 0) = 0.37$–0.42) and increased to 0.02–0.11 under high precipitation ($\Pr(\Delta > 0) = 0.70$–0.84; Table S-3). In fenced plots, positive effects appeared only at the highest precipitation and were weaker. *Poa autumnalis* showed little survival benefit in unfenced plots across the gradient (median $\Delta = -0.018$ to -0.002, $\Pr(\Delta > 0) = 0.14$–0.43), but weakly positive effects emerged when herbivores were excluded (median $\Delta = 0.002$–0.002, $\Pr(\Delta > 0) = 0.57$–0.67).

Growth responses increased with precipitation across species, with the magnitude and certainty modulated by herbivory (Table S-4). In *A. hyemalis*, effects were strongly positive even at moderate precipitation in unfenced plots (median $\Delta = 0.37$–0.39; $\Pr(\Delta > 0) = 0.95$–0.99), and similar trends were observed in fenced plots though slightly weaker ($\Delta = 0.08$–0.26; $\Pr(\Delta > 0) = 0.66$–0.87). *E. virginicus* showed small positive or neutral effects under dry conditions (median $\Delta = 0.23$–0.26; $\Pr(\Delta > 0) = 0.87$–0.95), which became more pronounced at higher precipitation. In *P. autumnalis*, growth was context-dependent: unfenced plots shifted from negative to positive effects across the precipitation gradient (median $\Delta = -0.28$ to 0.14;

$\Pr(\Delta > 0) = 0.17–0.77$), while fencing generally amplified benefits at all precipitation levels (median $\Delta = 0.02–0.39$; $\Pr(\Delta > 0) = 0.56–0.99$).

Epichloë endophytes also increased flowering and spikelet production under wetter conditions, though effects were species- and herbivore-dependent (Figs. 4, S-4; Figs. S-5). Flowering benefits in *A. hyemalis* and *E. virginicus* were minimal under dry conditions but increased with precipitation, with minor influence from herbivore exclusion. In contrast, *P. autumnalis* showed strong flowering benefits under fencing at moderate-to-high precipitation (median $\Delta = 0.47–1.54$; $\Pr(\Delta > 0) = 0.99$; Table S-5). Spikelet production followed a similar pattern: *E. virginicus* and *P. autumnalis* showed weak or negative effects under low precipitation, shifting to positive effects with higher precipitation, and fencing amplified these benefits across the gradient (median $\Delta = 0.10–1.56$; $\Pr(\Delta > 0) = 0.77–0.99$; Table S-6).

Abiotic factors as the primary drivers of fitness components across the stress gradient

Grass-endophyte symbiosis, whether mutualistic or parasitic, was primarily modulated by climatic variation rather than by herbivore pressure. Across all vital rates (survival, growth, inflorescence, and spikelet production), models comparing different drivers of the endophyte–host interaction consistently identified the *Endophyte* $\times$ *Climate* model as the best predictor of host demography, based on expected log predictive density (ELPD; higher values indicate better fit; Table 1). For instance, for survival, the *Endophyte* $\times$ *Climate* model outperformed the *Endophyte* $\times$ *Herbivory* model (ΔELPD $= -2.1$, SE = 2.8), and including the three-way *Endophyte* $\times$ *Climate* $\times$ *Herbivory* interaction further decreased predictive performance (ΔELPD $= -5.0$, SE = 1.9). Similar patterns were observed for growth, inflorescence, and spikelet production, indicating that climatic context is the primary driver of variation in the demographic consequences of endophyte symbiosis, whereas herbivory has weaker or context-dependent effects.

Table 1: Model comparison of drivers of grass- endophyte symbiosis across four fitness components (vital rates). ΔELPD indicates the difference in expected log predictive density relative to the best-fitting model; SE is the standard error of the difference.

Vital rate	Model type	ΔELPD	SE
Survival	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−2.11	2.81
	Endophyte × Climate × Herbivory	−4.98	1.89
Growth	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.19	2.81
	Endophyte × Climate × Herbivory	−1.81	2.75
Inflorescence	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−2.62	3.53
	Endophyte × Climate × Herbivory	−3.78	2.63
Spikelet	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−2.01	3.82
	Endophyte × Climate × Herbivory	−3.16	2.43

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Figures

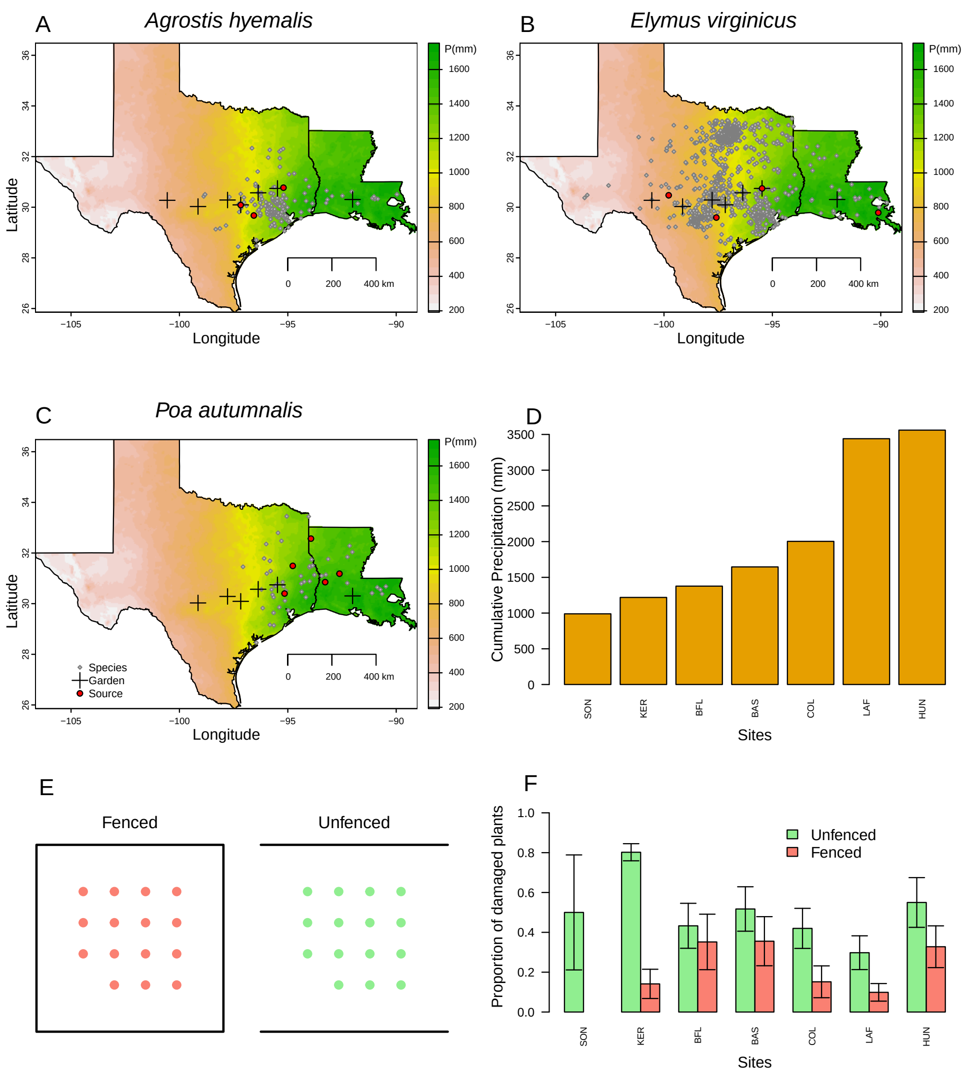

Figure 1: Distribution of common garden sites along a natural aridity gradient from Texas to Louisiana, US for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Climate data in A–C are based on 30-year normals of climate predictions (1993–2023) from PRISM data (Hart and Bell, 2015), which illustrate the precipitation gradient across sites (scale on the right is in mm. Red dots indicate the locations of source populations, while grey dots represent Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area (GBIF, 2025a,b,c). D) Cumulative precipitation during the study period (May 2023 – June 2025) at each site, ordered west to east. Values are also derived from PRISM climate data (Hart and Bell, 2015).

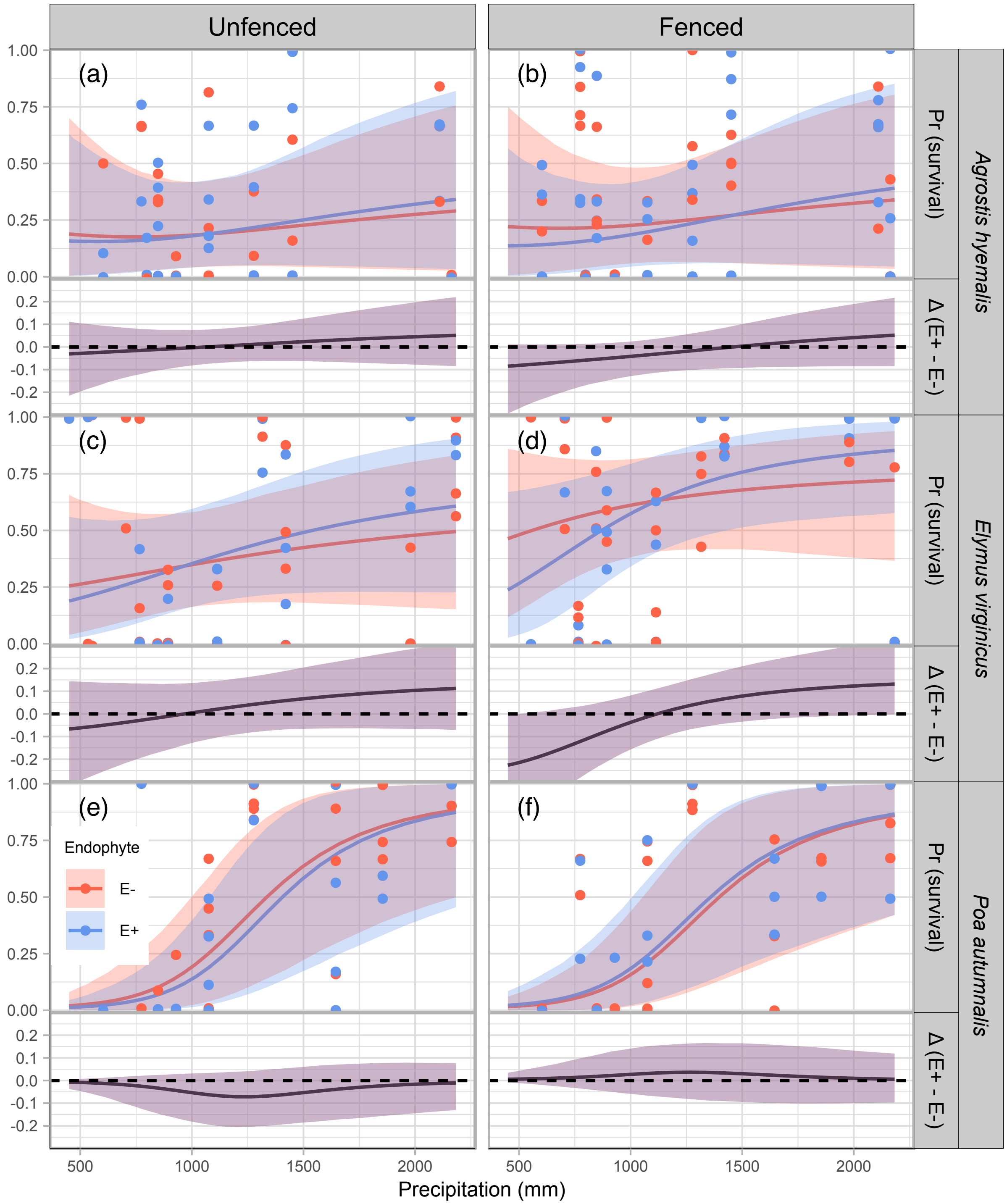

Figure 2: Predicted survival rate (top panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed survival per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

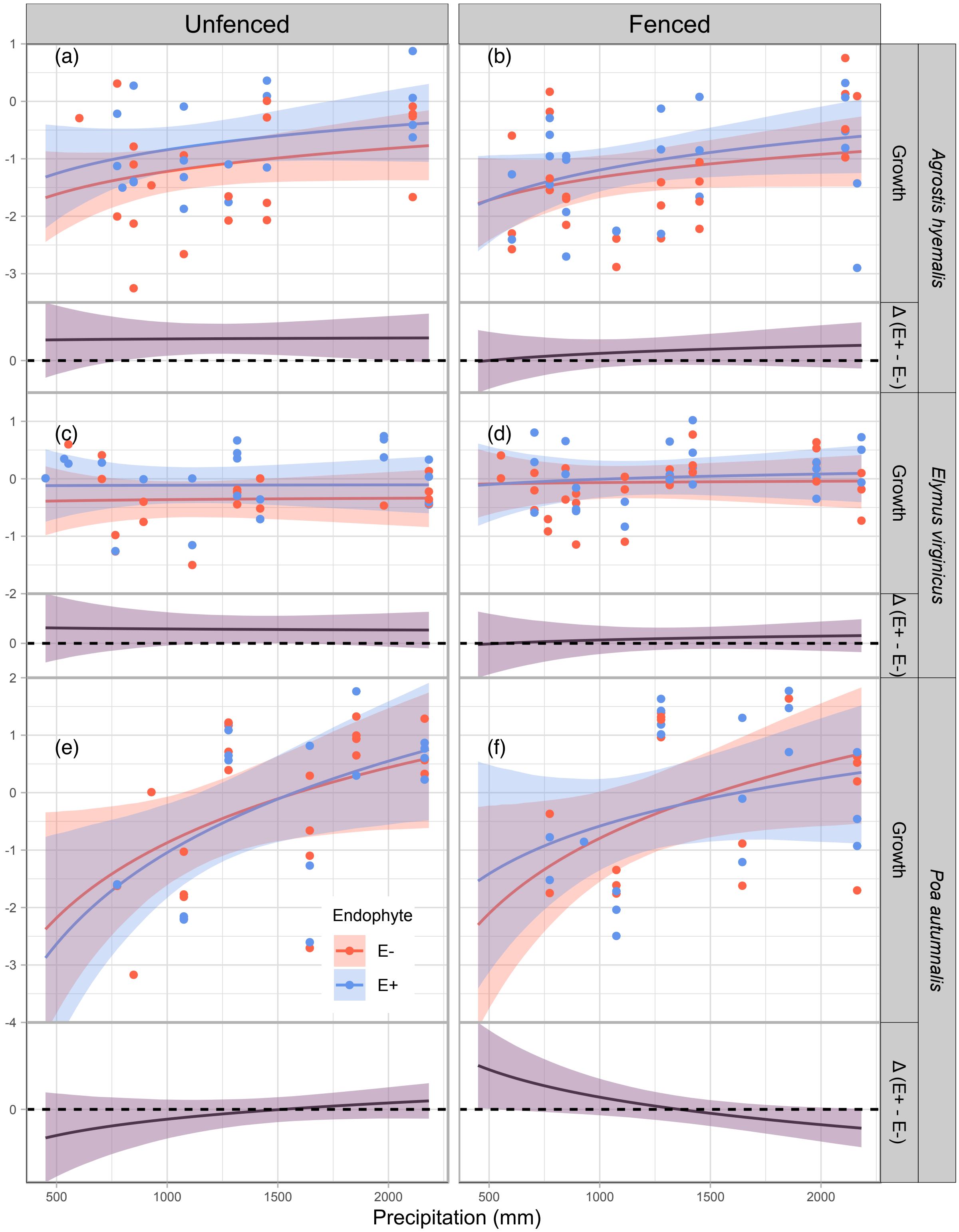

Figure 3: Predicted relative growth from year to the next year (top panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed growth per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

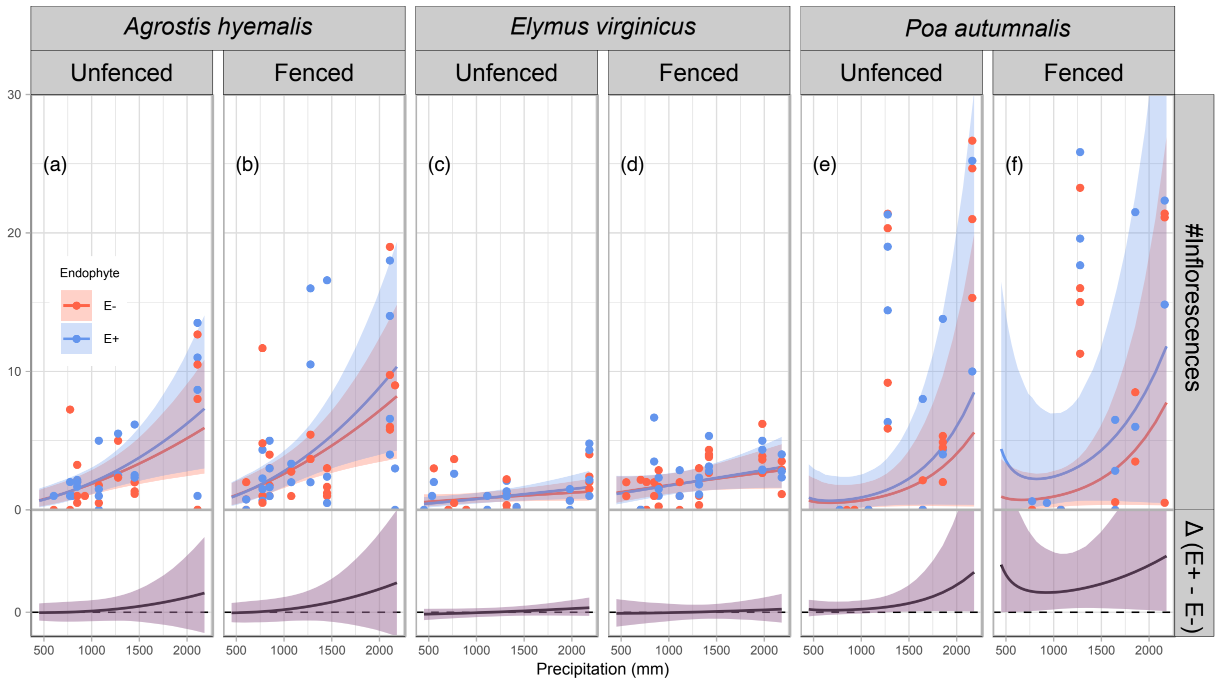

Figure 4: Predicted number of inflorescences (upper panels) and the difference between endophyte-infected (E^+) and endophyte-free (E^-) plants ($\Delta(E^+–E^-)$, lower panels) are shown across a gradient of precipitation. Shaded areas represent 90% credible intervals. Points indicate observed means,jittered for visualization. Panels are grouped by species and herbivory treatment (unfenced = with herbivory; fenced = herbivory excluded).

S.1 Supporting Information

List of Supplementary Figures

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S.1.1 Supporting Methods

S.1.2 Candidate models for abiotic and biotic drivers of cool-season grass species

1. **Endophyte × Climate model:** includes endophyte status (endo), precipitation (P), and their interaction:

$$\mu_i = \beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_P[\mathrm{Sp}_i] P_i + \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i] endo_i P_i$$
$$+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$$

2. **Endophyte × Herbivory model:** includes endophyte status (endo), herbivory (herb), and their interaction:

$$\mu_i = \beta_0[\mathrm{Sp}_i] + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i] herb_i + \beta_{\mathrm{endo}\times \mathrm{herb}}[\mathrm{Sp}_i] endo_i herb_i$$
$$+ \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$$

3. **Endophyte × Climate × Herbivory:** combines all predictors and interactions up to the three-way interaction:

$$\mu_i = \beta_0[\mathrm{Sp}_i] + \beta_P[\mathrm{Sp}_i] P_i + \beta_{\mathrm{endo}}[\mathrm{Sp}_i] endo_i + \beta_{\mathrm{herb}}[\mathrm{Sp}_i] herb_i$$
$$+ \beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i] endo_i P_i + \beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i] herb_i P_i + \beta_{\mathrm{herb}\times \mathrm{endo}}[\mathrm{Sp}_i] herb_i endo_i$$
$$+ \beta_{\mathrm{endo}\times \mathrm{herb}\times P}[\mathrm{Sp}_i] endo_i herb_i P_i + \phi_{\mathrm{site}[i],\mathrm{Sp}_i} + \omega_{\mathrm{source}[i],\mathrm{Sp}_i} + \rho_{\mathrm{plot}[i],\mathrm{Sp}_i}$$

In these models, the β coefficients represent species-specific effects. The intercept $\beta_0[\mathrm{Sp}_i]$ corresponds to the baseline value of the vital rate for species Sp_i in the absence of predictors. $\beta_{\mathrm{endo}}[\mathrm{Sp}_i]$ and $\beta_{\mathrm{herb}}[\mathrm{Sp}_i]$ describe the effects of endophyte presence and herbivory, respectively. The linear effect of precipitation is captured by $\beta_P[\mathrm{Sp}_i]$, while interactions are represented by $\beta_{\mathrm{endo}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{herb}\times P}[\mathrm{Sp}_i]$, $\beta_{\mathrm{endo}\times \mathrm{herb}}[\mathrm{Sp}_i]$, and the three-way interaction $\beta_{\mathrm{endo}\times \mathrm{herb}\times P}[\mathrm{Sp}_i]$. All models include hierarchical random effects for site-year, source population, and plot, with species-specific random effects $\phi_{\mathrm{site}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{site}})$, $\omega_{\mathrm{source}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{source}})$, and $\rho_{\mathrm{plot}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0,\sigma_{\mathrm{plot}})$.

S.1.3 Supporting Tables

Supptable S1: Geographic coordinates and brief descriptions of source populations for *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (POAU).

Species	Population	Latitude	Longitude	Description
AGHY	SJC	30.771969	-95.200620	Ecolab property
AGHY	COHR	29.672833	-96.567517	Columbus, TX herbarium record
AGHY	LPEF	30.085383	-97.172508	Lost Pines
POAU	LA7	30.849660	-93.276772	DeRidder, LA
POAU	LA1	32.568125	-93.932976	Jean Lafitte National Historical Park and Preserve
POAU	LA2	31.183040	-92.616969	Jean Lafitte National Historical Park and Preserve
POAU	SFAEF	31.492158	-94.772200	Stephen F. Austin Experimental Forest, TX
POAU	WB	30.399346	-95.150026	Sam Houston National Forest
ELVI	HUNT	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	SHS	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	RL5	30.471717	-99.783803	Texas Tech University, Junction
ELVI	PALM	29.591623	-97.584781	Palmetto State Park, TX
ELVI	JLP	29.785105	-90.114933	Jean Lafitte Park, outside of New Orleans

Table S-1: Common garden and field experiment sites used for reciprocal transplant and environmental manipulation studies.

Site	Code	Latitude	Longitude
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
1335 Medina Hwy, Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S-2: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOOCV). ELPD difference is relative to the best model.

Vital rate	Model type	ELPD difference	SE difference
Survival	Quadratic	0	0.000
Survival	Linear	-1.230	0.947
Growth	Quadratic	0	0.000
Growth	Linear	-0.544	0.924
Flowering	Quadratic	0	0.000
Flowering	Linear	-0.763	0.956
Spikelet	Quadratic	0	0.000
Spikelet	Linear	-0.097	0.957

Table S-3: Posterior summaries of Δ ($E^{+} - E^{-}$) for survival at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
Agrostis hyemalis	Unfenced	765.76	-0.01055	-0.12081	0.07964	0.3723
Agrostis hyemalis	Unfenced	848.28	-0.00814	-0.10368	0.07597	0.3980
Agrostis hyemalis	Unfenced	1113.25	0.00173	-0.07149	0.07904	0.5239
Agrostis hyemalis	Unfenced	1644.36	0.02230	-0.06632	0.14693	0.6981
Agrostis hyemalis	Unfenced	2163.28	0.03508	-0.08410	0.21801	0.7237
Agrostis hyemalis	Fenced	765.76	-0.04897	-0.17775	0.01302	0.0961
Agrostis hyemalis	Fenced	848.28	-0.04477	-0.15803	0.01516	0.1076
Agrostis hyemalis	Fenced	1113.25	-0.02642	-0.11372	0.03630	0.2266
Agrostis hyemalis	Fenced	1644.36	0.01141	-0.08847	0.12851	0.5950
Agrostis hyemalis	Fenced	2163.28	0.03789	-0.08484	0.21422	0.7222
Elymus virginicus	Unfenced	765.76	-0.02574	-0.19555	0.13388	0.3799
Elymus virginicus	Unfenced	848.28	-0.01565	-0.17000	0.13230	0.4250
Elymus virginicus	Unfenced	1113.25	0.01937	-0.10368	0.14589	0.6070
Elymus virginicus	Unfenced	1644.36	0.07749	-0.06409	0.22970	0.8217
Elymus virginicus	Unfenced	2163.28	0.10622	-0.07003	0.31076	0.8445
Elymus virginicus	Fenced	765.76	-0.12411	-0.28674	0.03254	0.0992
Elymus virginicus	Fenced	848.28	-0.09198	-0.23593	0.04709	0.1418
Elymus virginicus	Fenced	1113.25	-0.00179	-0.11213	0.11041	0.4882
Elymus virginicus	Fenced	1644.36	0.08855	-0.02165	0.24020	0.9069
Elymus virginicus	Fenced	2163.28	0.11327	-0.00261	0.32611	0.9456
Poa autumnalis	Unfenced	765.76	-0.00979	-0.12238	0.02097	0.1853
Poa autumnalis	Unfenced	848.28	-0.01778	-0.14683	0.02450	0.1735
Poa autumnalis	Unfenced	1113.25	-0.05636	-0.20042	0.03503	0.1429
Poa autumnalis	Unfenced	1644.36	-0.02891	-0.17290	0.07018	0.2804
Poa autumnalis	Unfenced	2163.28	-0.00174	-0.13201	0.07681	0.4319
Poa autumnalis	Fenced	765.76	0.00399	-0.03466	0.09583	0.6685
Poa autumnalis	Fenced	848.28	0.00739	-0.04282	0.11317	0.6803
Poa autumnalis	Fenced	1113.25	0.02557	-0.06906	0.15496	0.7063
Poa autumnalis	Fenced	1644.36	0.01678	-0.09915	0.15764	0.6297
Poa autumnalis	Fenced	2163.28	0.00181	-0.09727	0.12122	0.5663

Table S-4: Posterior summaries of Δ ($E^+ - E^-$) for growth across precipitation quantiles for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes; negative medians indicate costly effects.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Agrostis hyemalis	Unfenced	773.91	0.37006	-0.00924	0.75437	0.9456
Agrostis hyemalis	Unfenced	1075.09	0.37940	0.10078	0.65111	0.9866
Agrostis hyemalis	Unfenced	1316.25	0.38541	0.11668	0.64408	0.9915
Agrostis hyemalis	Unfenced	1979.35	0.39167	0.01393	0.77022	0.9539
Agrostis hyemalis	Unfenced	2163.28	0.39436	-0.02748	0.81310	0.9400
Agrostis hyemalis	Fenced	773.91	0.07955	-0.23563	0.39683	0.6612
Agrostis hyemalis	Fenced	1075.09	0.13909	-0.10277	0.37262	0.8279
Agrostis hyemalis	Fenced	1316.25	0.17455	-0.07319	0.41710	0.8781
Agrostis hyemalis	Fenced	1979.35	0.24676	-0.11528	0.60922	0.8733
Agrostis hyemalis	Fenced	2163.28	0.26153	-0.13527	0.66118	0.8657
Elymus virginicus	Unfenced	773.91	0.25869	-0.11938	0.62862	0.8693
Elymus virginicus	Unfenced	1075.09	0.24977	-0.02146	0.51978	0.9343
Elymus virginicus	Unfenced	1316.25	0.24421	0.00389	0.48413	0.9530
Elymus virginicus	Unfenced	1979.35	0.23379	-0.05653	0.52131	0.9096
Elymus virginicus	Unfenced	2163.28	0.23175	-0.08415	0.54666	0.8877
Elymus virginicus	Fenced	773.91	0.03164	-0.30420	0.37363	0.5644
Elymus virginicus	Fenced	1075.09	0.06458	-0.16044	0.29000	0.6825
Elymus virginicus	Fenced	1316.25	0.08512	-0.10777	0.27375	0.7682
Elymus virginicus	Fenced	1979.35	0.12472	-0.12987	0.37637	0.7903
Elymus virginicus	Fenced	2163.28	0.13352	-0.15046	0.41561	0.7788
Poa autumnalis	Unfenced	773.91	-0.27505	-0.74845	0.20422	0.1688
Poa autumnalis	Unfenced	1075.09	-0.14041	-0.44570	0.16567	0.2237
Poa autumnalis	Unfenced	1316.25	-0.05910	-0.29128	0.17786	0.3413
Poa autumnalis	Unfenced	1979.35	0.10761	-0.15864	0.37505	0.7388
Poa autumnalis	Unfenced	2163.28	0.14420	-0.16050	0.44856	0.7749
Poa autumnalis	Fenced	773.91	0.39049	-0.05133	0.83526	0.9270
Poa autumnalis	Fenced	1075.09	0.16142	-0.12313	0.45470	0.8212
Poa autumnalis	Fenced	1316.25	0.02003	-0.20891	0.25959	0.5587
Poa autumnalis	Fenced	1979.35	-0.26222	-0.54759	0.03589	0.0723
Poa autumnalis	Fenced	2163.28	-0.32245	-0.64896	0.01200	0.0558

Table S-5: Posterior summaries of Δ ($E^+ - E^-$) for flowering output at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
Agrostis hyemalis	Unfenced	773.91	-0.6824	-0.6904	0.7822	0.4889
Agrostis hyemalis	Unfenced	1075.09	0.0286	-0.6483	0.9444	0.6018
Agrostis hyemalis	Unfenced	1316.25	0.4663	-0.6218	1.2633	0.6909
Agrostis hyemalis	Unfenced	1979.35	1.3488	-1.2041	4.0378	0.7552
Agrostis hyemalis	Unfenced	2163.28	1.5409	-1.4924	5.3973	0.7522
Agrostis hyemalis	Fenced	773.91	-0.6824	-0.7351	0.9077	0.5238
Agrostis hyemalis	Fenced	1075.09	0.0286	-0.6302	1.2286	0.6696
Agrostis hyemalis	Fenced	1316.25	0.4663	-0.6108	1.7763	0.7609
Agrostis hyemalis	Fenced	1979.35	1.3488	-1.3208	5.6835	0.7965
Agrostis hyemalis	Fenced	2163.28	1.5409	-1.7235	7.3637	0.7922
Elymus virginicus	Unfenced	773.91	-0.6824	-0.4386	0.2723	0.3289
Elymus virginicus	Unfenced	1075.09	0.0286	-0.3390	0.3120	0.4611
Elymus virginicus	Unfenced	1316.25	0.4663	-0.2792	0.3903	0.5915
Elymus virginicus	Unfenced	1979.35	1.3488	-0.2491	0.8798	0.7902
Elymus virginicus	Unfenced	2163.28	1.5409	-0.2647	1.0691	0.8082
Elymus virginicus	Fenced	773.91	-0.6824	-0.8003	0.7325	0.4283
Elymus virginicus	Fenced	1075.09	0.0286	-0.6057	0.5997	0.4688
Elymus virginicus	Fenced	1316.25	0.4663	-0.4972	0.5628	0.5208
Elymus virginicus	Fenced	1979.35	1.3488	-0.6334	1.0660	0.6272
Elymus virginicus	Fenced	2163.28	1.5409	-0.7473	1.3341	0.6318
Poa autumnalis	Unfenced	773.91	-0.6824	-0.1587	0.7919	0.7062
Poa autumnalis	Unfenced	1075.09	0.0286	-0.0711	0.9392	0.8638
Poa autumnalis	Unfenced	1316.25	0.4663	0.0137	1.3447	0.9629
Poa autumnalis	Unfenced	1979.35	1.3488	0.0793	6.8120	0.9926
Poa autumnalis	Unfenced	2163.28	1.5409	0.0645	10.6951	0.9858
Poa autumnalis	Fenced	773.91	-0.6824	0.0907	5.6618	0.9981
Poa autumnalis	Fenced	1075.09	0.0286	0.2492	4.3096	0.9998
Poa autumnalis	Fenced	1316.25	0.4663	0.3164	4.7790	0.9999
Poa autumnalis	Fenced	1979.35	1.3488	0.1612	11.8419	0.9976
Poa autumnalis	Fenced	2163.28	1.5409	0.0888	15.3801	0.9848

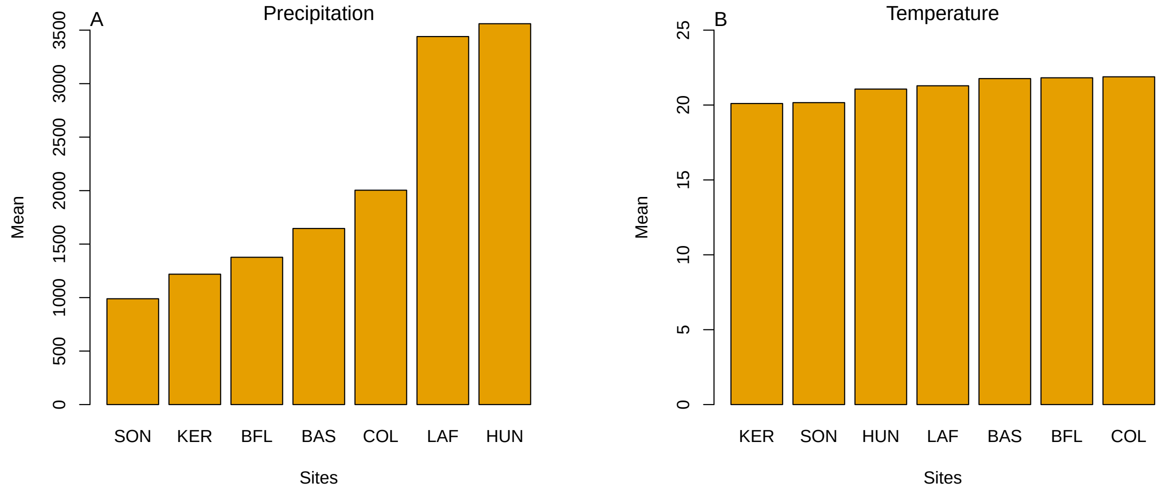
Table S-6: Posterior summaries of Δ ($E^{+} - E^{-}$) at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
Elymus virginicus	Unfenced	1113.25	0.1040	-4.4921	2.4788	0.3213
Elymus virginicus	Unfenced	1276.99	0.4008	-3.6975	2.7993	0.4012
Elymus virginicus	Unfenced	1644.36	0.9477	-2.5664	3.8884	0.6089
Elymus virginicus	Unfenced	2163.28	1.5409	-2.0652	6.1715	0.7733
Elymus virginicus	Unfenced	2182.71	1.5603	-2.0610	6.2851	0.7760
Elymus virginicus	Fenced	1113.25	0.1040	-4.2244	1.7020	0.2288
Elymus virginicus	Fenced	1276.99	0.4008	-3.0122	2.0977	0.3714
Elymus virginicus	Fenced	1644.36	0.9477	-1.2591	3.8036	0.7853
Elymus virginicus	Fenced	2163.28	1.5409	-0.4058	7.9946	0.9282
Elymus virginicus	Fenced	2182.71	1.5603	-0.3886	8.1667	0.9288
Poa autumnalis	Unfenced	1113.25	0.1040	-3.2874	1.7757	0.3134
Poa autumnalis	Unfenced	1276.99	0.4008	-2.7023	1.8008	0.3654
Poa autumnalis	Unfenced	1644.36	0.9477	-1.7415	2.3862	0.5832
Poa autumnalis	Unfenced	2163.28	1.5409	-1.9577	5.4441	0.7572
Poa autumnalis	Unfenced	2182.71	1.5603	-1.9811	5.5917	0.7592
Poa autumnalis	Fenced	1113.25	0.1040	0.3203	6.9989	0.9668
Poa autumnalis	Fenced	1276.99	0.4008	0.6863	6.6041	0.9828
Poa autumnalis	Fenced	1644.36	0.9477	0.8105	6.7216	0.9873
Poa autumnalis	Fenced	2163.28	1.5409	-0.9449	8.6686	0.8957
Poa autumnalis	Fenced	2182.71	1.5603	-1.0369	8.7690	0.8906

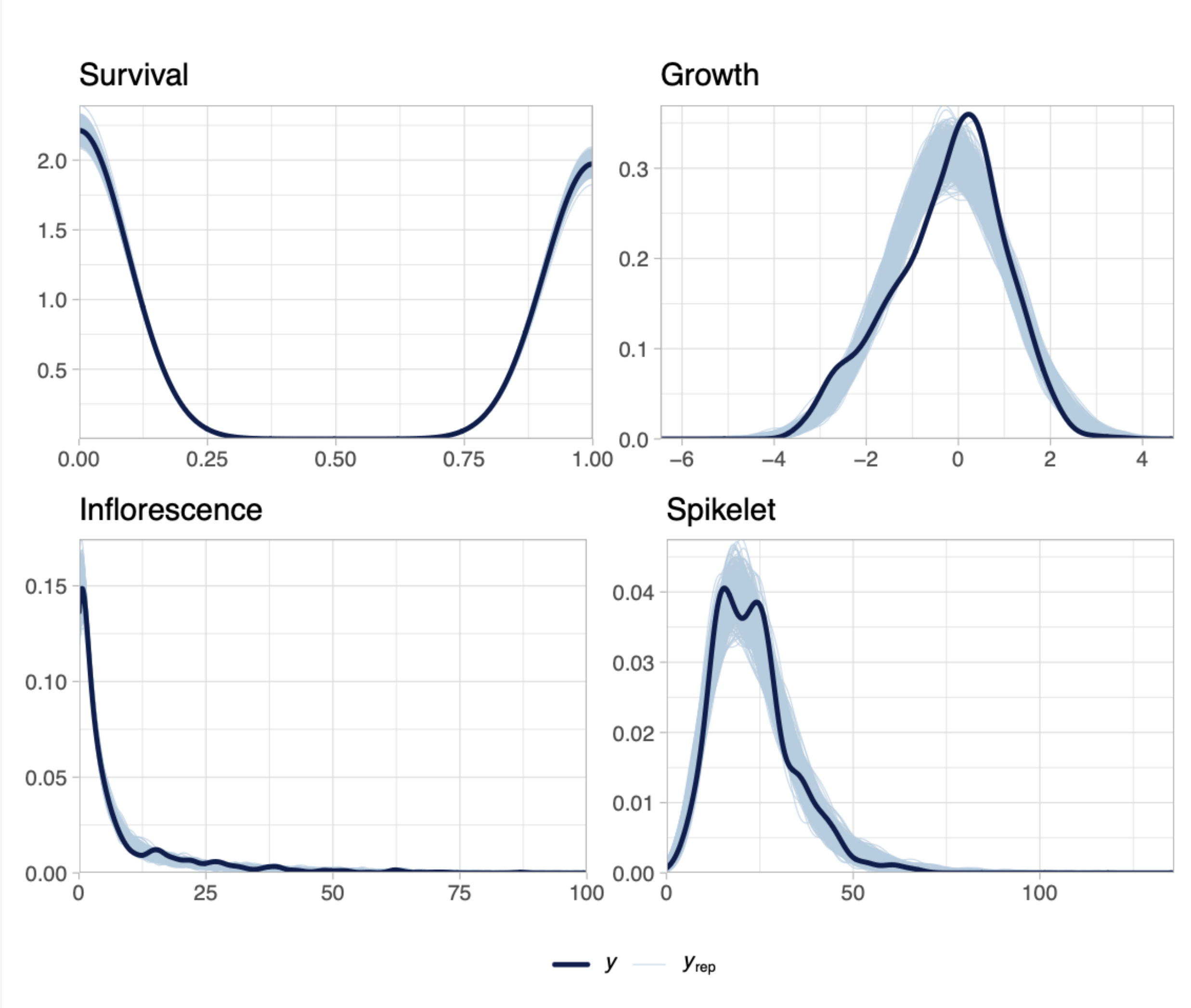
Table S-7: Posterior summaries of interaction effects (mean estimate, 95% credible intervals, and posterior probability) for survival, growth, flowering, and spikelet production across three grass species.

Fitness component	Species	Parameter	Estimate	Lower CI	Upper CI	Probability
Survival	*Agrostis hyemalis*	Endophyte × Climate	1.213	0.658	2.354	0.730
Survival	*Elymus virginicus*	Endophyte × Climate	1.364	0.666	2.802	0.806
Survival	*Poa autumnalis*	Endophyte × Climate	1.255	0.483	3.344	0.681
Survival	*Agrostis hyemalis*	Endophyte × Herbivory	0.804	0.364	1.720	0.287
Survival	*Elymus virginicus*	Endophyte × Herbivory	0.868	0.347	2.119	0.376
Survival	*Poa autumnalis*	Endophyte × Herbivory	1.941	0.642	6.026	0.879
Survival	*Agrostis hyemalis*	Endophyte × Herbivory × Climate	1.149	0.503	2.525	0.630
Survival	*Elymus virginicus*	Endophyte × Herbivory × Climate	1.450	0.560	3.880	0.774
Survival	*Poa autumnalis*	Endophyte × Herbivory × Climate	0.737	0.229	2.394	0.303
Growth	*Agrostis hyemalis*	Endophyte × Climate	-0.005	-0.340	0.329	0.488
Growth	*Elymus virginicus*	Endophyte × Climate	-0.013	-0.290	0.268	0.461
Growth	*Poa autumnalis*	Endophyte × Climate	0.208	-0.156	0.559	0.873
Growth	*Agrostis hyemalis*	Endophyte × Herbivory	-0.283	-0.734	0.158	0.107
Growth	*Elymus virginicus*	Endophyte × Herbivory	-0.208	-0.621	0.218	0.169
Growth	*Poa autumnalis*	Endophyte × Herbivory	0.365	-0.135	0.862	0.929
Growth	*Agrostis hyemalis*	Endophyte × Herbivory × Climate	0.084	-0.296	0.462	0.666
Growth	*Elymus virginicus*	Endophyte × Herbivory × Climate	0.060	-0.296	0.405	0.634
Growth	*Poa autumnalis*	Endophyte × Herbivory × Climate	-0.549	-0.966	-0.141	0.004
Flowering	*Agrostis hyemalis*	bendoclim	1.092	0.711	1.646	0.656
Flowering	*Elymus virginicus*	bendoclim	1.178	0.817	1.698	0.805
Flowering	*Poa autumnalis*	bendoclim	1.103	0.725	1.689	0.676
Flowering	*Agrostis hyemalis*	bendoherb	1.014	0.577	1.784	0.523
Flowering	*Elymus virginicus*	bendoherb	0.987	0.549	1.781	0.478
Flowering	*Poa autumnalis*	bendoherb	1.964	1.052	3.807	0.982
Flowering	*Agrostis hyemalis*	bendoherbclim	0.997	0.604	1.670	0.496
Flowering	*Elymus virginicus*	bendoherbclim	0.900	0.567	1.437	0.332
Flowering	*Poa autumnalis*	bendoherbclim	0.640	0.380	1.055	0.038
Spikelet	*Elymus virginicus*	bendoclim	1.100	0.930	1.310	0.866
Spikelet	*Poa autumnalis*	bendoclim	1.103	0.888	1.362	0.817
Spikelet	*Elymus virginicus*	bendoherb	1.029	0.760	1.395	0.573
Spikelet	*Poa autumnalis*	bendoherb	1.315	0.969	1.786	0.961
Spikelet	*Elymus virginicus*	bendoherbclim	1.023	0.806	1.293	0.573
Spikelet	*Poa autumnalis*	bendoherbclim	0.873	0.682	1.119	0.145

S.1.4 Supporting Figures



Suppfigure S1: Abiotic conditions across the seven study sites. (A) Mean annual precipitation showing the pronounced east–west aridity gradient from mesic Louisiana to arid Texas. (B) Mean annual temperature across sites, which does not differ much. Panels highlight that precipitation, rather than temperature, constitutes the primary abiotic gradient in this system.



Suppfigure S2: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y), and light blue lines show replicated datasets generated from posterior predictions (y_{rep}). The strong overlap across survival, growth, flowering, and spikelet production indicates adequate model fit for all vital rates.

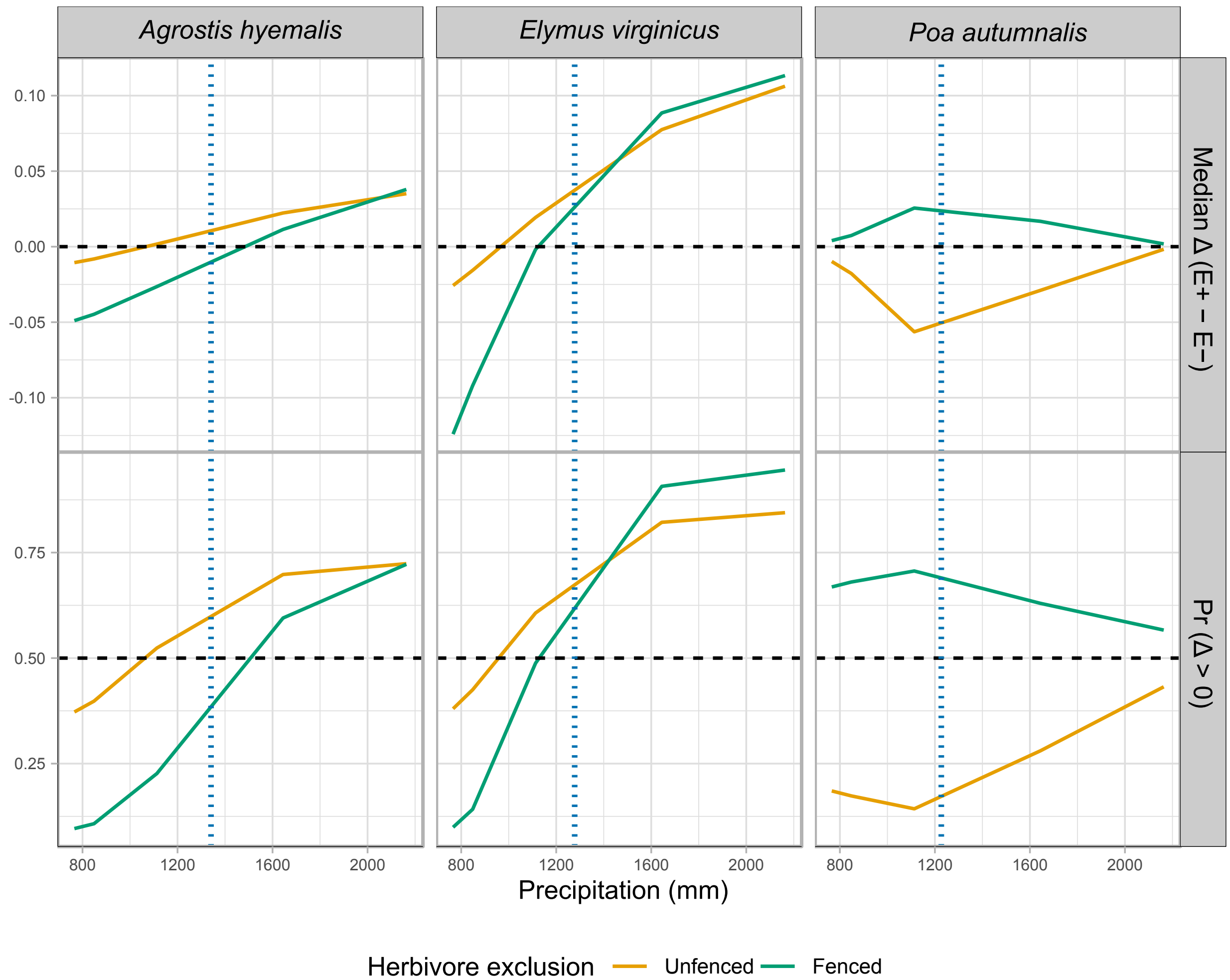


Figure S-1: Predicted effects of endophyte presence (E^+) versus absence (E^-) on survival across precipitation gradients for three cool-season grasses. Solid lines show the median Δ ($E^+ - E^-$),or the posterior probability that $\Delta > 0$ under fenced and unfenced conditions. Horizontal dashed lines indicate no effect ($\Delta = 0$) or a 50% probability of benefit. Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

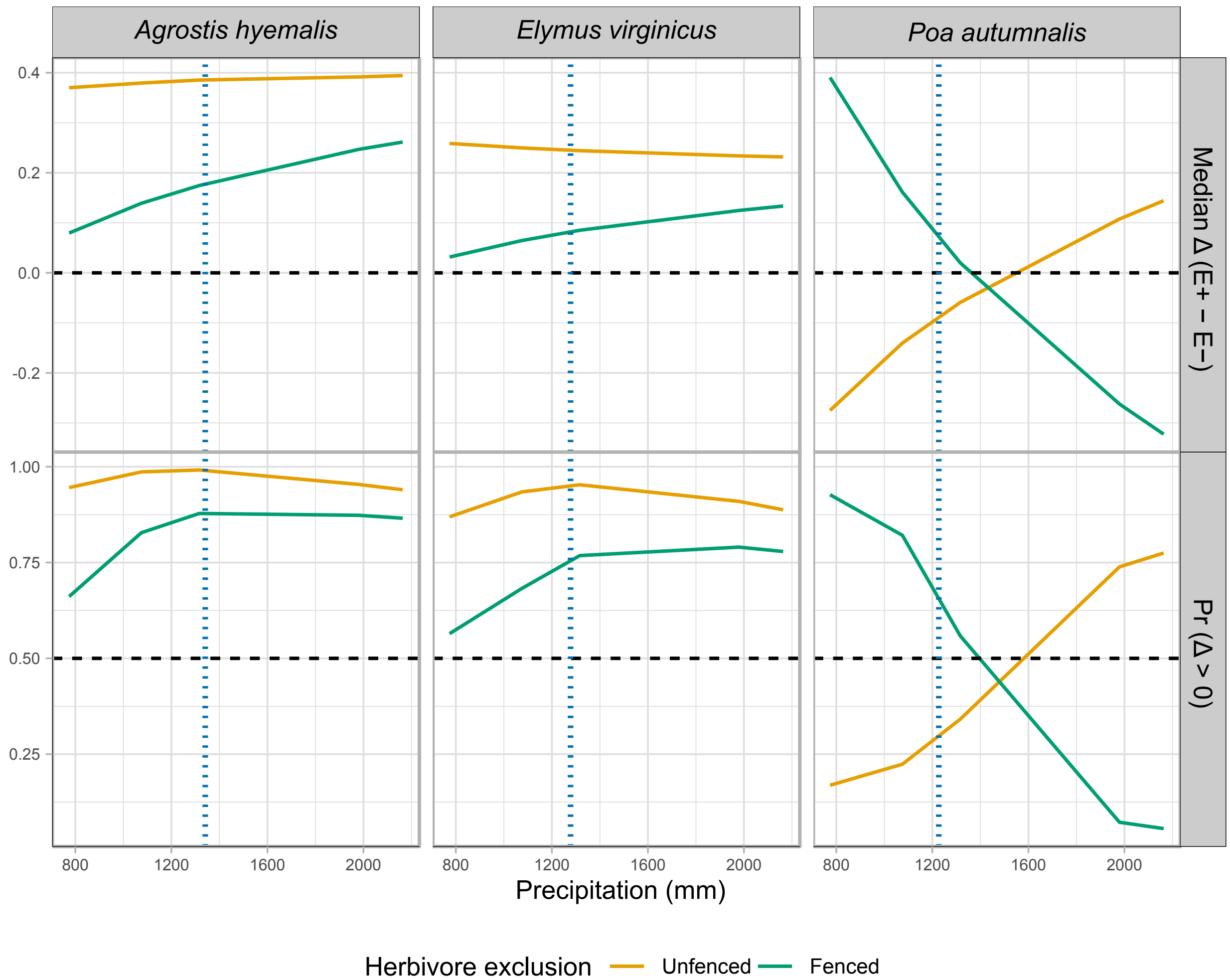


Figure S-2: Effects of precipitation and endophyte presence on growth across three cool-season grass species. Lines show the predicted difference in growth between endophyte-present (E^+) and endophyte-absent (E^-) plants (Median Δ, top row) and the posterior probability that this difference is positive ($\Pr(\Delta > 0)$, bottom row) under fenced and unfenced treatments. Predictions were calculated at the 10th, 25th, 50th, 75th, and 90th percentiles of precipitation observed in the study. Horizontal dashed lines indicate $\Delta = 0$ (top row) and $\Pr(\Delta > 0) = 0.5$ (bottom row). Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

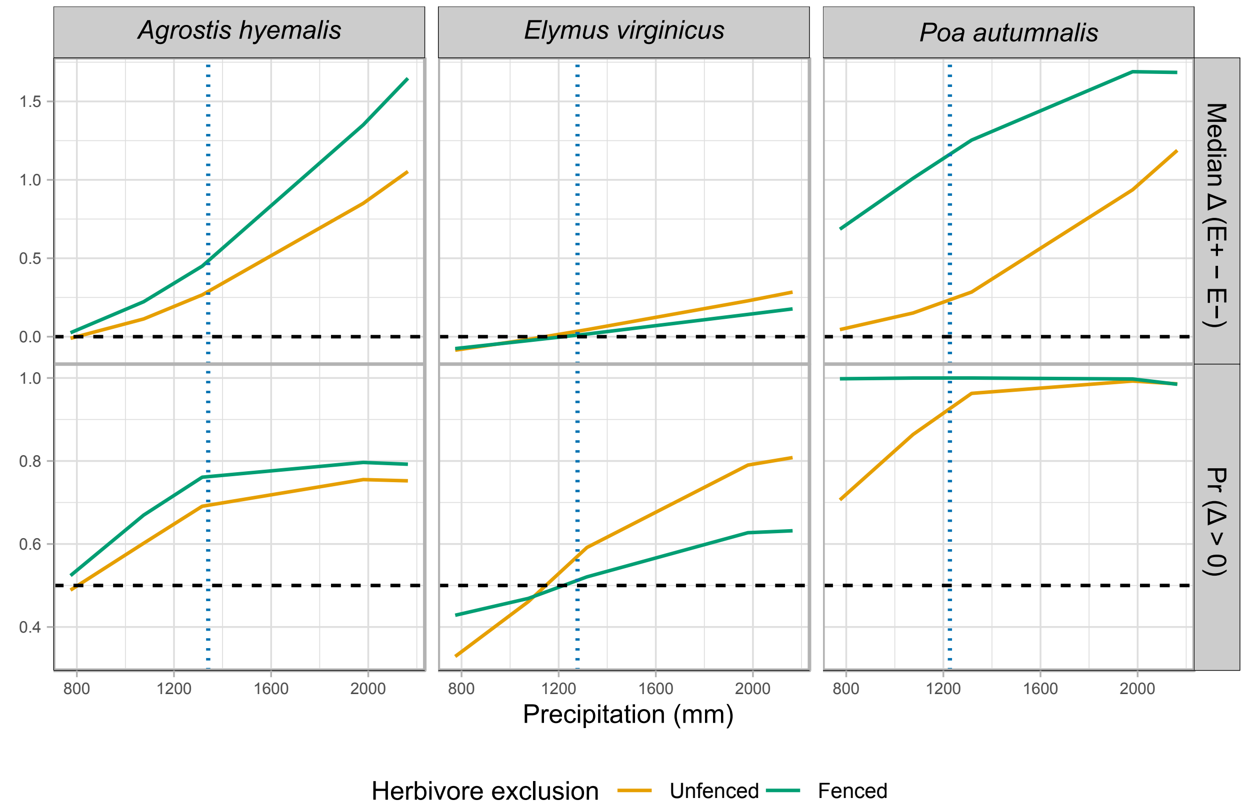


Figure S-3: Effects of endophyte infection on flowering output across a precipitation gradient for three cool-season grass species. Lines show the median posterior difference Δ ($E^{+} - E^{-}$) and the posterior probability that $\Delta > 0$ under fenced (herbivores excluded) and unfenced (herbivores present) conditions. Horizontal dashed lines indicate $\Delta = 0$ for median differences and $\Pr(\Delta > 0) = 0.5$. Endophyte effects were generally positive for *Agrostis hyemalis* and *Poa autumnalis*, but for *Elymus virginicus* benefits were context-dependent, with costs at lower precipitation

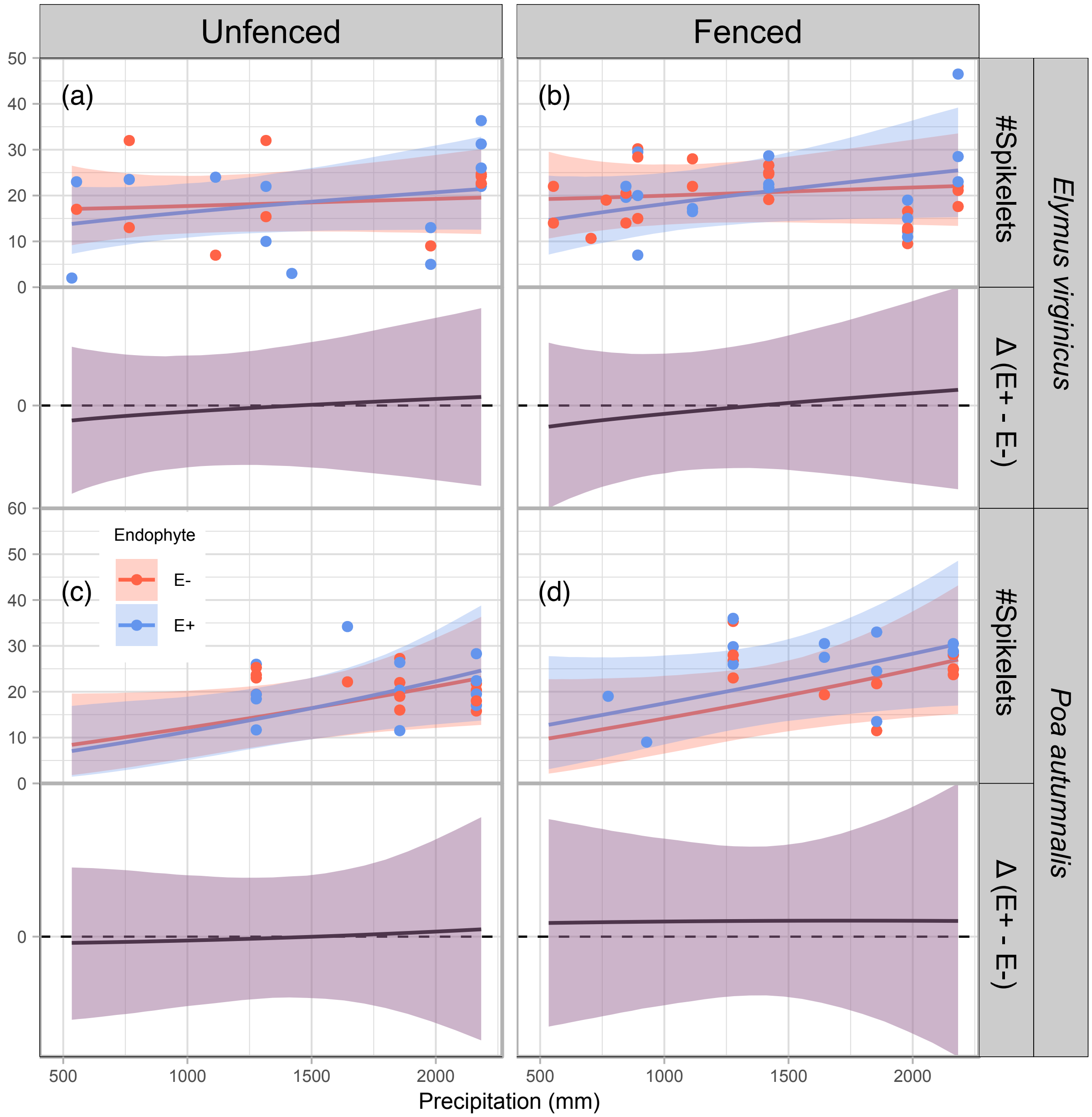


Figure S-4: Predicted number of spikelets (upper panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for only two species (see Methods for details). The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

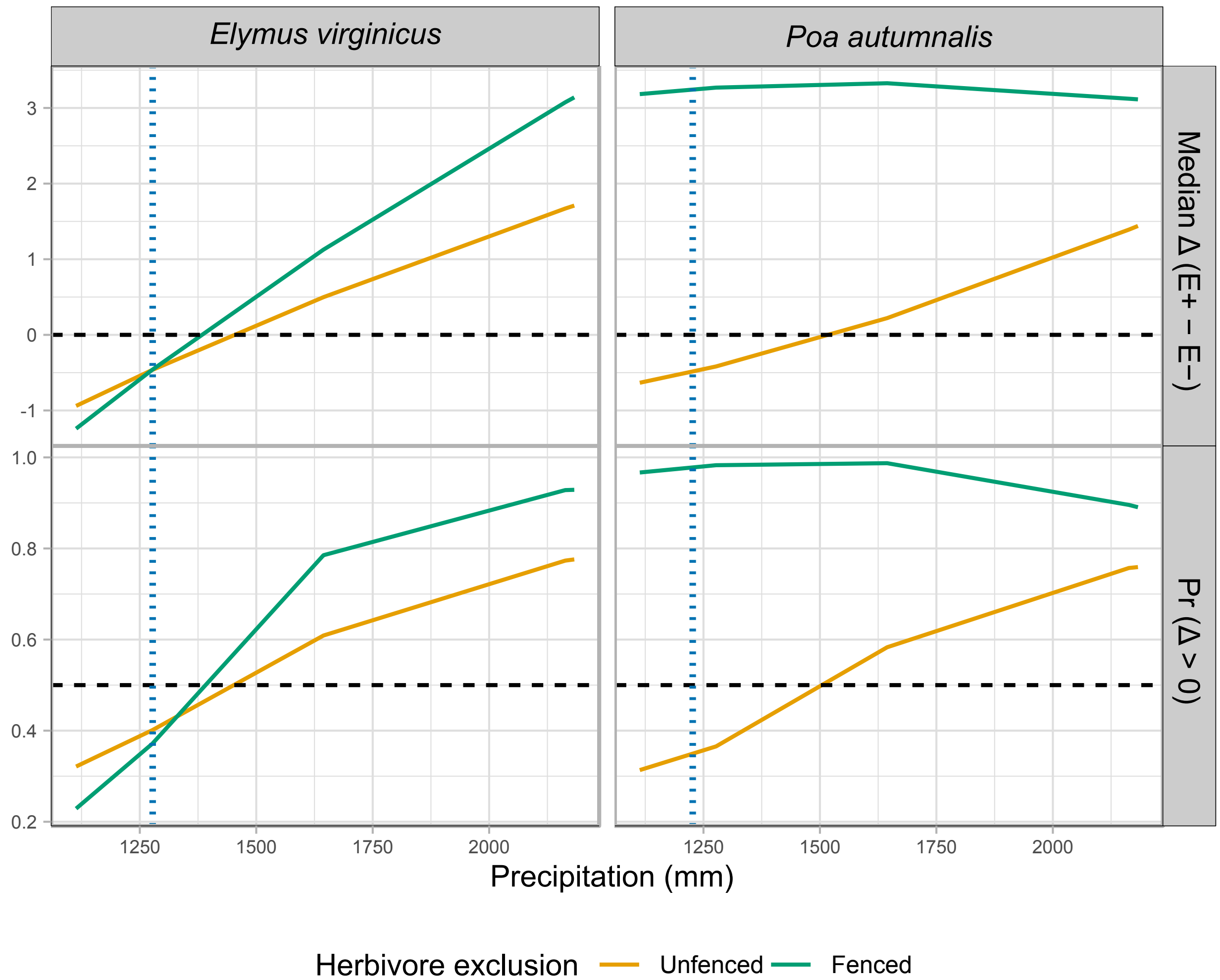


Figure S-5: Effects of precipitation and endophyte presence on spike (reproductive output) for three cool-season grass species. Median Δ ($E^{+} - E^{-}$) and posterior probability $Pr(\Delta > 0)$ are shown across precipitation quantiles. Herbivore exclusion treatments are indicated by color (Fenced = herbivores excluded, Unfenced = herbivores had access). Horizontal dashed lines indicate $\Delta = 0$ for medians and $Pr(\Delta > 0) = 0.5$ for probabilities. Facets separate metrics and species.